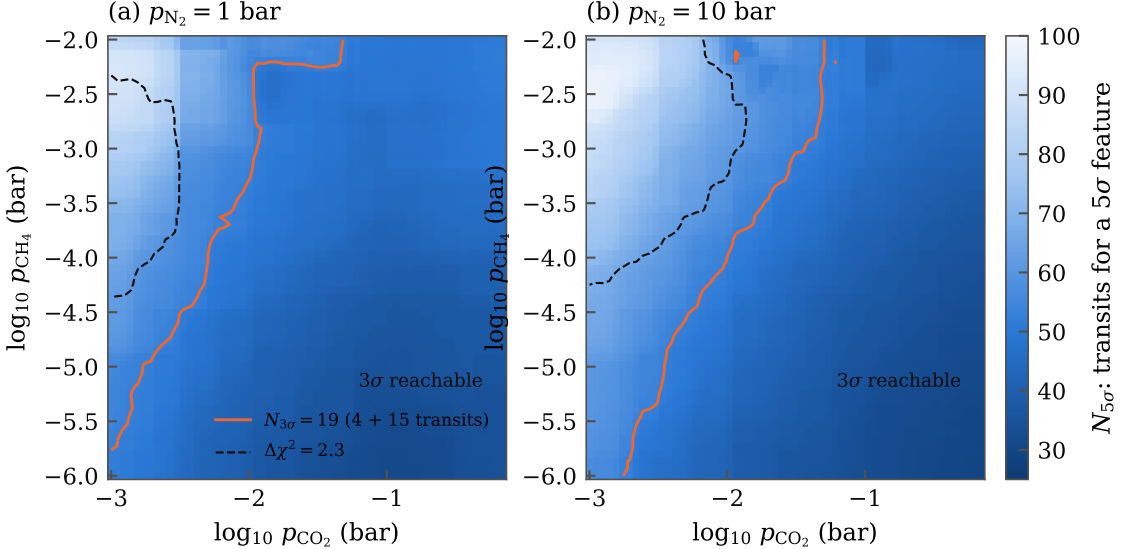
(a) $p_{N_2} = 1$ bar
(b) $p_{N_2} = 10$ bar
$\log_{10} p_{CH_4}$ (bar)
$\log_{10} p_{CO_2}$ (bar)
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5
−5.0
−5.5
−6.0
−3
−2
−1
3σ reachable
$N_{3\sigma} = 19$ (4 + 15 transits)
$\Delta\chi^2 = 2.3$
100
90
80
70
60
50
40
30
$N_{5\sigma}$: transits for a 5σ feature